MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 22: Friday 30 October 2026

(Week 11)

Remark. Q21.3 to Q21.5 are leftover exercises from the Tuesday class.

Remark. Q22.1 is additional exercise on triple integrals in spherical coordinates (Section 14.7).

Q21.3. Evaluate the iterated integral

$$\int_{\pi/6}^{\pi/3} \int_{\csc\phi}^{2\csc\phi} \int_{0}^{2\pi} \rho^2 \sin\phi \, d\theta \, d\rho \, d\phi.$$

Q21.4. Evaluate the following integral:

$$\int_{-1}^{1} \int_{0}^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} \sqrt{x^2+y^2+z^2} \; dz \, dy \, dx.$$

Q21.5. Determine the volume of the region in the xyz-space bounded below by the cone $z^2 = x^2 + y^2$ and bounded above by the sphere of radius 1 centered at the origin.

Q22.1. Consider the solid region E in the xyz-space consisting of points (x, y, z) such that $1 \leq z \leq 2$ and $x^2 + y^2 + z^2 \leq 4$. Use spherical coordinates to evaluate $\iiint_E (x^2 + y^2 + z^2)^{-1/2}\, dV$.